

ICOHP Descriptors and Thermodynamic Stability

Across a 349-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

Gilles Frapper
 Professor, Theoretical Chemistry
 IC2MP UMR 7285 CNRS, Université de Poitiers
gilles.frapper@univ-poitiers.fr

Report generated as part of the `viability` project

August 16, 2026

Update, August 16, 2026: this report is reorganized around the project’s current central question — $\Delta(\text{ICOHP})$ and viability discrimination (§13) — rather than around the percolation weight, the original campaign 2 subject. Everything concerning the percolation weight and its direct descendants (network dimensionality, periodic min-cut — missions #1 and #3) has been moved to Appendix A, “Percolation_viability,” with the corresponding source code relocated to the repository’s `Percolation_viability/` directory. The main body below documents the project’s current state (349 structures, missions #4–#5, the `extension4` campaign, the $\Delta(\text{ICOHP})$ reframing) without altering the results themselves.

Contents

1	Objective and Problem Statement	3
2	Methodology	3
2.1	Dataset Selection	3
2.2	Bonding-Type Classification	3
2.3	VASP + LOBSTER Computation	3
2.4	Statistics	4
3	Workflow Diagram	4
4	Environment	4
5	Dataset	5
6	Results	5
6.1	Correlations (Spearman) — classic ICOHP aggregates	5
7	Failures Found and Fixed	6
8	Compute Cost	6
9	Current Limits and Next Steps	6
10	Conclusion	7

11 Campaign 2 follow-ups (missions #4 and #5)	7
11.1 Mission #4: antibonding population near the Fermi level	8
11.2 Mission #5: reaction ICOHP	8
11.3 Comparison (missions #4–#5, 186/192-compound scale)	8
12 August 16 Update: the extension4 Campaign and Missions #4–#5 Repro-	8
cessed	
12.1 The extension4 campaign	8
12.2 Missions #4–#5, reprocessed at 349 compounds	9
13 Project Reframing (August 16): the Central Question is Δ(ICOHP)-Based	11
Viability	
13.1 Reference concordance, formalized	11
13.2 Extended to every compound computed to date, pooled without regard to campaign	12
13.3 Reading	13
A Appendix: Percolation_viability	13
A.1 Original objective (mission #1)	13
A.2 Original workflow diagram	14
A.3 Campaign 2 dataset (186 compounds)	14
A.4 Results (with the percolation weight)	15
A.5 Mission #3: network dimensionality and periodic min-cut	16
A.6 Limits specific to the percolation weight	17
A.7 Conclusion of the original campaign	18
A.8 Compound List (percolation weight)	18
B VASP Calculation Parameters	22
C PAW-PBE Potentials	22
D k-point Grid and NBANDS per Compound	24
E Extension Campaign: Elements and Allotropes	27
F Extension Campaign: Compounds and Polymorphs	29
G Extension Campaign: Endobondic Reactions	31
H Extension Campaign: Exobondic Reactions	32

1 Objective and Problem Statement

Central research question (see §13): does $\Delta(\text{ICOHP})$ — the reaction-ICOHP descriptor for a compound’s decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

This question was built up progressively. The first report (6 toy compounds) established that a percolation descriptor diverges sharply from classic ICOHP aggregates, but without a sample large enough to statistically test whether that difference is *useful*. The original campaign 2 therefore built a 186-compound dataset to settle that question — work whose methodology (dataset selection, statistical conventions) remains the foundation everything below builds on, but whose descriptor itself (the percolation weight) and its own result are now documented in Appendix A, not in this report’s main body: its headline correlation did not survive the dataset’s growth (see §12), and it is no longer this project’s organizing question.

Three later missions (network dimensionality/periodic min-cut — #3, also in the appendix; antibonding population — #4; reaction ICOHP — #5) and an extension campaign of 163 additional compounds made $\Delta(\text{ICOHP})$ emerge as the project’s strongest and most robust correlation (§11–§12), before the central question was explicitly reformulated around it (§13).

2 Methodology

2.1 Dataset Selection

Materials Project query (`mp_dataset/select_campaign.py`) for the main dataset (186 compounds): unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP’s categorical `magnetic_ordering` field is "Unknown" for ~98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements, ≤ 20 atoms/cell, 60 compounds per family with a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system. Four further extension batches (163 structures, §12) apply the same non-magnetic/no-f filter but with their own per-mission selection logic (elemental references, high-pressure polymorphs, alkali/alkaline-earth binaries) — detailed in their respective sections, not repeated here.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by `fetch_candidates.py`’s `classify()` function (**reused unchanged**), from `is_metal` and elemental composition. **This classification is purely informative in this report:** a descriptive lens on the dataset (§5), not a sampling axis or a variable any main-body conclusion depends on — other mission reports (`REPORT_antibonding.md`, `REPORT_reaction_icohp.md`) actively use it to stratify their own correlations, but that is not its role here.

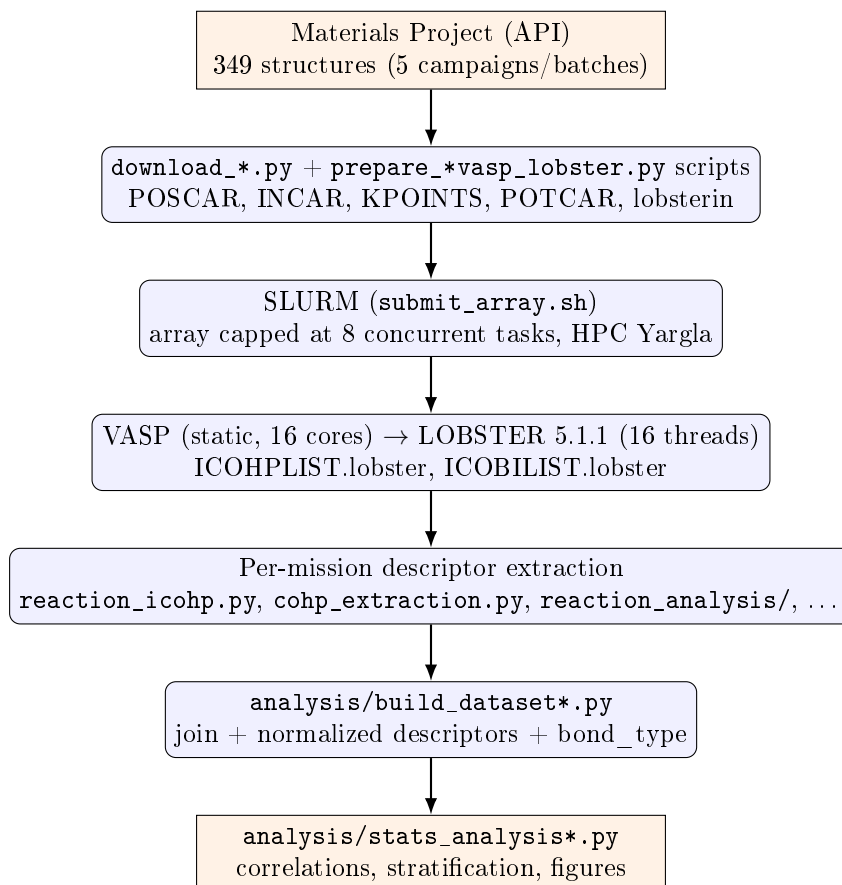
2.3 VASP + LOBSTER Computation

Same template across all 349 structures (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): NBANDS is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOB descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the mission), overall and per subgroup (`bond_type`, `is_metal`, or selection provenance as appropriate); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



The percolation weight (`Percolation_viability/percolation_path.py`) and its direct descendants (dimensionality, min-cut) follow the same pipeline upstream of the "descriptor extraction" step but no longer appear in this diagram, or as this report's subject — see the original workflow diagram and the detail of their own processing in Appendix A.

4 Environment

Identical across all 349 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991
extension (4 batches) [†]	163	heterogeneous, see §11–§12
Total	349	

[†] elemental references, TiO₂/carbon high-pressure polymorphs, and alkali/alkaline-earth binaries (**extension4**) — full composition in Appendix E–F.

Composition	n
Single-element references	68
Multi-element compounds	281
Total	349

Bond type (classify() , informative only)	n
metallic	88
ionic	53
covalent	35
unclassified	173
Total	349

The "bond type" table was **not out of date by accident**: the **extension4** campaign (§12) specifically added 89 alkali/alkaline-earth binaries against N/O/F/P/S/Cl, taking the ionic total from 9 (campaign 2 alone) to 53. Half of these new binaries nonetheless remain "unclassified": **classify()** only recognizes alkali/alkaline-earth $\times$ {halogen, O} pairs as "ionic" (**IONIC_ANIONS**), not $\times$ {N, P, S} — a real limitation of the reused heuristic, documented but not fixed here. As noted in §2.2, this categorization remains purely descriptive in this report.

6 Results

6.1 Correlations (Spearman) — classic ICOHP aggregates

Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

These four rows (classic ICOHP aggregates, campaign 2, $n = 186$, target **energy_above_hull**) are kept here as the reference baseline for every later mission. **The percolation weight, which appeared in this same table in the original version of this report, is moved in full to Appendix A** (complete table with bond-type subgroups, reference logistic regression, and associated figures) — this trimmed table therefore no longer allows a direct percolation-vs-classic-aggregates comparison; that comparison is still available as-is in the appendix. Current descriptors' correlations (antibonding population, reaction ICOHP) are given in §11–§12, against **formation_energy_per_atom** rather than **energy_above_hull**.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the 178 SLURM-array-submitted jobs failed on the first pass while computing campaign 2's 186 compounds, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to every later campaign):

- **Al12W, BW, WO2, TeW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **Al12W** failed again after that fix, this time with a VASP SIGSEGV — exactly the known issue flagged in the mission brief (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 349 structures computed to date: **349/349**.

8 Compute Cost

349 structures computed in total: 68 single-element references and 281 multi-element compounds (§5), across five submission campaigns/batches (the original campaign 2 and four extension batches).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1, srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The SLURM array (`submit_array.sh`) is capped at 8 concurrent tasks.

	VASP	LOBSTER
<i>n</i>	349	348*
Mean (s)	168	404
Median (s)	58	110
Max (s)	5715 (1.6 h)	12063 (3.4 h)
Sequential total	16.3 h	39.0 h

* 1 of 349 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 349 structures): **55.3 h**. Given the 8-concurrent-task cap, the order of magnitude for actual elapsed time to process this volume as a single batch would be ~ 6.9 h — a rough estimate (dividing by 8, ignoring queue time) given for context: in practice, the 349 structures were submitted across several campaigns spread over multiple sessions, not as one single batch.

9 Current Limits and Next Steps

- **classify() coverage:** 173/349 structures (50%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth $\times$ {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any mission that would want to stratify on it.

- **No true dynamical "unstable" class:** `theo_metastable` and the theoretical `extension` batches are proxies (never synthesized), not a phonon-calculation result.
- **Per-subgroup sample size:** remains this project's structural limit at every new stratification (`bond_type`, `is_metal`, `family`) — see §11–§12 for concrete examples where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet:** reaction ICOHP (mission #5, §12) is now the strongest correlate and the first whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified feature search, still not attempted.

Limits specific to the percolation weight and its descendants (primitive-vs-conventional-cell sensitivity, chance-level reference logistic regression) are documented in Appendix A, not repeated here.

10 Conclusion

The dataset grew from 186 to 349 computed structures (VASP+LOBSTER) across five campaigns/batches, with a final success rate of 349/349 and two compute failures diagnosed and fixed at the source during the original campaign 2. The project's organizing question evolved along the way: starting from the percolation weight (no significant correlation retained, Appendix A), through network dimensionality/periodic min-cut (also in the appendix, headline correlation not reproduced at the current scale), the antibonding population near the Fermi level (mission #4, the project's most robust covalent subgroup), to reaction ICOHP and its endobondic/exobondic classification (mission #5) — now the project's strongest and most robust correlation, and the central question explicitly reformulated around it (§13). The next concrete milestone, already identified in the reframing, is a deliberate extension to compounds near the stability edge rather than continuing to accumulate compounds that are comfortably stable or comfortably unstable.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_formation_energy.csv`

Detailed reports: `analysis/REPORT_*.md`

11 Campaign 2 follow-ups (missions #4 and #5)

Two further descriptors were built in later sessions, both tested against `formation_energy_per_atom` (introduced in mission #3 as an alternative target to `energy_above_hull`, generally more correlatable) rather than the hull. Both reuse the 186-compound dataset above unmodified at the initial stage; mission #5 draws in part on a **74-compound extension campaign** (`extension_*`: elemental references, ionic compounds and polymorphs outside the main set), detailed in §12. Numbers in this subsection are as of August 15, 2026, at the 186/192-compound scale; §12 reprocesses both missions at the 349-compound scale reached the following day and should be read alongside this subsection, not instead of it.

A pattern repeats across both missions: each new descriptor reaches a statistically significant *global* correlation, stronger than anything the percolation weight or the min-cut ever achieved (Appendix A) — but in both cases that global correlation turns out to be substantially, or entirely, a clustering effect between bond types or between metals and gapped compounds, rather than a robust relationship *within* a homogeneous group.

11.1 Mission #4: antibonding population near the Fermi level

A distinct question from integrated ICOHP (one number per bond): not *how much* bonding a compound has, but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller electronic instabilities (new module `cohp_extraction.py`, validated by exact cross-check against `ICOHPLIST.lobster` on the 6 pilot compounds). Headline: $\rho = -0.328$, $p = 5.0 \times 10^{-6}$ ($n = 186$) — the strongest global correlation in the project at the time. The sign is *opposite* the naive Peierls/Jahn–Teller reading. Diagnostics: the effect is substantially driven by metal/gapped clustering, but the `bond_type=covalent` subgroup ($n = 23$) survives stratification ($p = 0.018$ – 0.029) — the only subgroup in the project, to date, to hold up under such a test.

11.2 Mission #5: reaction ICOHP

An ICOHP analog of `formation_energy_per_atom`: is a compound's total ICOHP "cheaper" than the same atoms in elemental reference form (new module `reaction_icohp.py`, reaction balanced via `pymatgen.analysis.reaction_calculator`)? Extended to 192 compounds using the 74-compound extension campaign (62 elemental references computed). Headline: $\rho = 0.370$, $p = 3.9 \times 10^{-7}$ ($n = 177$) — numerically the strongest global correlation in the project to date, this time with an intuitive sign (less bonding than the elemental state $\Rightarrow$ less stable formation). But **no bond_type subgroup survives stratification** (Kruskal–Wallis $p = 2.1 \times 10^{-6}$ for the descriptor, $p = 5.0 \times 10^{-8}$ for the target, across the three strata) — unlike mission #4, the clustering effect here explains the entire observed signal. Against `energy_above_hull` the correlation is absent ($\rho = 0.09$, $p = 0.20$).

11.3 Comparison (missions #4–#5, 186/192-compound scale)

On the same common subsample ($n = 177$, the intersection of campaign 2's 186 compounds and mission #5's 192 computable rows):

Descriptor	n	ρ	p
Reaction ICOHP (mission #5)	177	0.370	3.9×10^{-7}
Antibonding population, normalized (mission #4)	177	-0.337	4.5×10^{-6}

Periodic min-cut (mission #3, $\rho = 0.313$) and percolation weight (mission #1, $\rho = 0.104$), on the same $n = 177$, are given for reference in Appendix A.

Ranking by raw correlation strength (ρ) is not the same as ranking by reliability: mission #5 leads numerically but is also the only one where *no* subgroup survives stratification, whereas mission #4 keeps a significant covalent subgroup.

12 August 16 Update: the extension4 Campaign and Missions #4–#5 Reprocessed

12.1 The extension4 campaign

89 new compounds (`mp_dataset/download_extension4.py`): binary compounds of an alkali (Li/Na/K/Rb/Cs) or alkaline-earth (Be/Mg/Ca/Sr/Ba) metal against N/O/F/P/S/Cl, sourced from Materials Project. Up to 2 compounds picked per (cation, anion) chemical system across 44 systems: an experimentally-known polymorph (preferring formulas already elsewhere in the dataset, to build genuine same-composition polymorph groups — the dataset had only 8 such groups before this batch) and the highest-`energy_above_hull` theoretical entry available for that formula, a deliberate far-from-equilibrium stress test. Selection was computed once and frozen to

`extension4_candidates.json` for reproducibility (scale made hand-curation, the convention for every earlier extension batch, impractical). All 89 VASP+LOBSTER jobs completed unattended (a self-healing watcher restarted any job that left the SLURM queue without producing output; zero restarts were actually needed). Dataset: 260 $\rightarrow$ 349 compounds.

Separately, the schema-driven `reaction_analysis/` package (superseding the ad hoc `reaction_icohp.py` module long-term) was extended with an **endobondic/exobondic bonding-kinetics classifier**: a reaction’s ΔICOHP (products – reactants) being positive ("endobondic") means breaking the reactant’s bonds costs more than the products’ bonds recover — a kinetic barrier that can keep an exothermically-favored decomposition from actually happening, explaining why some very exothermic compounds are nonetheless observed to exist. Validated against 7 reference reactions with independently known ICOHP values; not yet connected to this project’s own LOBSTER data for classification, an explicit next step, not attempted this session.

Every reaction’s per-bond-type ICOHP values were transcribed from independent reference values (`tests/fixtures/reitz_dronskowski_cases.yaml`), independent of this project’s own CSP data, then run end to end through `balance.py/delta.py/classify.py`. All 7 match the reference ΔICOHP within a few kJ/mol (attributable to the source’s own rounding of its reported per-bond eV values) and every `BondingLabel` matches:

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label	
			Ref.	Computed
$\text{Pb}(\text{N}_3)_2 \rightarrow \text{Pb} + 3 \text{N}_2$	+1345	+1344.3	endo	endo
$\text{S}_4\text{N}_2 \rightarrow \frac{1}{2}\text{S}_8 + \text{N}_2$	+258	+258.5	endo	endo
$\text{S}_4\text{N}_4 \rightarrow \frac{1}{2}\text{S}_8 + 2\text{N}_2$	+399	+397.3	endo	endo
$\text{ZnSn} \rightarrow \text{Zn} + \text{Sn}$	−337	−336.6	exo	exo
$\text{CaO}[\text{sphalerite}] \rightarrow \text{CaO}[\text{rocksalt}]$	−79	−78.9	exo	exo
$\text{CaN} \rightarrow \frac{1}{3}\text{Ca}_3\text{N}_2 + \frac{1}{6}\text{N}_2$	−205	−204.8	exo	exo
$\text{Mn}_2\text{O}_7 \rightarrow 2\text{MnO}_2 + \frac{3}{2}\text{O}_2$	−186	−187.7	exo	exo

The Mn_2O_7 case is a useful guard-rail on its own: exobondic by this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than by sudden bond rupture — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than hardcoding Mn_2O_7 as a special case.

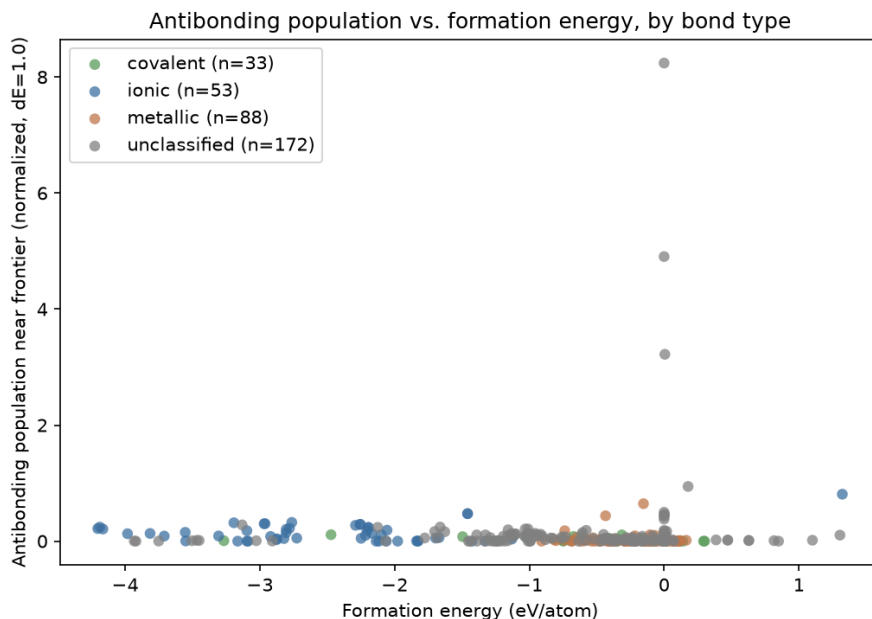
The full list of every compound and element computed across the entire **extension** campaign — not just **extension4**’s 89, but all four batches combined (163 compounds total: elemental references, carbon allotropes, TiO_2 high-pressure polymorphs, and every ionic/ covalent compound added along the way) — together with the case-1 decomposition reactions computed from them (their own ΔICOHP /atom and endobondic/exobondic label, distinct from the 7 reference cases above), is given in Appendix E through Appendix H.

12.2 Missions #4–#5, reprocessed at 349 compounds

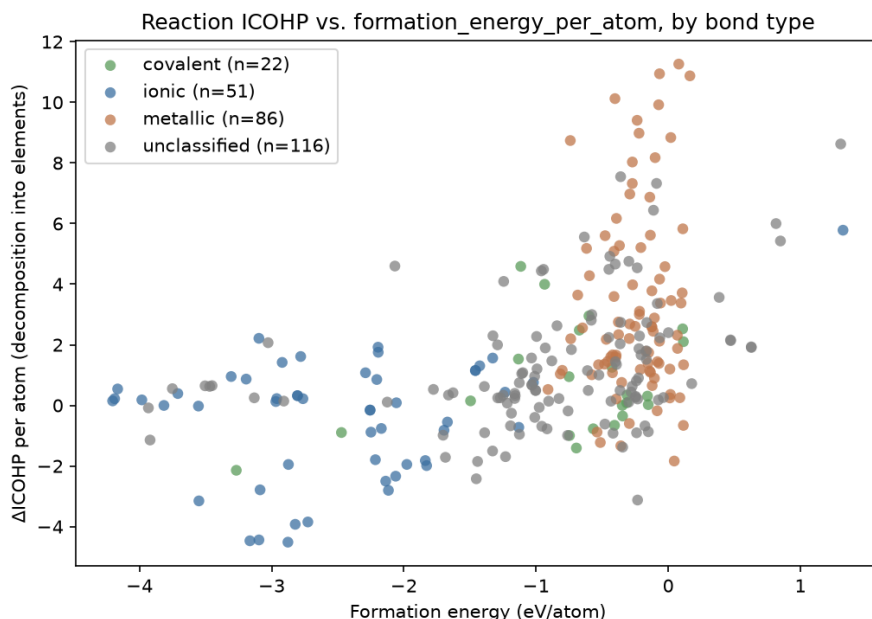
Every per-compound descriptor script was rerun unmodified over the grown dataset — **extension4**’s constituent elements were all already covered by the existing elemental-reference set, so no new element calculations were needed. (Mission #3, network dimensionality/periodic min-cut, was also reprocessed at this scale; its own result, and the diagnosis of its headline correlation’s collapse, are documented in Appendix A, not here.)

Mission #4 (antibonding): $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$) — weaker than the original $\rho = -0.328$, but the `bond_type=covalent` subgroup got *stronger* ($n = 33$, $\rho = -0.551$, $p = 0.0009$, vs. the original $\rho = -0.490$ at $n = 23$) and remains the single most robust stratified result in the project. `is_metal=False` newly survives ($n = 157$, $p = 0.001$). The old

`bond_type=ionic` result ($n = 9$, $\rho = -0.683$) did not replicate once the sample grew to $n = 53$ ($\rho = -0.184$, not significant) — a clean demonstration of why $n < 15$ subgroups are flagged rather than trusted.



Mission #5 (reaction ICOHP): $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — up from $\rho = 0.370$, and now the clear strongest correlation in the project by a wider margin than before. `bond_type` stratification still fully explains that layer of the pooled signal (Kruskal–Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but `is_metal` stratification now survives on both sides** (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$), where at $n = 186$ neither did. This revises the previous verdict that "the between-group confound explains the entirety of the observable signal" — it explains the `bond_type`-level pooling completely, but not a real, newly-detectable within-`is_metal`-group relationship. Case 2 (polymorph comparison), extended from 8 to **49 groups** specifically because `extension4` was designed to fix that sampling gap: 53.1% agreement between most-bonding and most-stable member, an exact Poisson-binomial test against the 45.9% chance baseline gives $P = 0.19$ — still no signal, but now a real test instead of an 8-point exploratory pass.



Still no SISSO. Reaction ICOHP is now the strongest raw correlate of `formation_energy_per_atom` in the project, and the first descriptor whose signal is not fully explained away by between-group clustering at every level tested — the strongest argument yet for a stratified (`is_metal-conditional`) feature search over a pooled one, still not attempted.

Detailed reports: `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

The diagnosis of mission #3's (min-cut) collapse is in Appendix A (`analysis/REPORT_dimensionality_mincut`).

13 Project Reframing (August 16): the Central Question is $\Delta(\text{ICOHP})$ -Based Viability

The project's central research question is restated as of this update: **does $\Delta(\text{ICOHP})$ — the reaction-ICOHP descriptor for a compound's decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?** This supersedes the percolation-weight-centric framing that organized the original campaign 2 (Appendix A) — percolation weight (mission #1) and network dimensionality/periodic min-cut (mission #3) remain valuable as the historical foundation (periodic graph construction, the statistical conventions every later mission still reuses) but are no longer this project's organizing question, consistent with min-cut's own headline correlation not surviving the dataset's growth to 349 compounds (§12, Appendix A). Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

13.1 Reference concordance, formalized

§12 already showed all 7 reference reactions reproducing to within a few kJ/mol. Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even one with a large systematic offset or scale factor. **Lin's Concordance Correlation Coefficient (CCC)** specifically tests agreement with the identity line (computed = reference), the actual claim being tested:

Statistic	Value
n	7
Lin’s CCC	0.999998
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean residual	0.76 kJ/mol
Max residual	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially within the reference values’ own reporting precision (rounded to 2–3 decimals), not merely that they move together — the implementation is validated by it, not just individually spot-checked.

13.2 Extended to every compound computed to date, pooled without regard to campaign

Deliberately not organized by which historical campaign or extension batch a compound came from — those boundaries reflect submission order across sessions, not chemistry, and grouping by them would itself be exactly the kind of between-group confound this project’s methodology exists to catch (the same lesson Appendix A’s min-cut diagnosis already demonstrated). **281 case-1 reactions**, every element and compound with a computed decomposition-into-elements reaction across the whole project, pooled into one population.

Continuous stability targets:

Target	n	ρ	p
formation_energy_per_atom	275	−0.5016	6.3×10^{-19}
energy_above_hull_eV_per_atom	279	−0.1096	0.068

(Sign: `reaction_analysis`’s products–reactants convention, opposite `reaction_icohp.py`’s own column reported elsewhere in this report — same magnitude, flipped sign.) The formation-energy result is the same relationship as §12’s mission-#5 headline, confirmed here on the pooled, campaign-agnostic population.

Sign-based (endobondic/exobondic) discrimination. Experimental- viability proxy (a compound’s `theoretical` flag: `False` means it has been experimentally realized):

	endobondic	exobondic
experimental ($n = 177$)	47	130
theoretical-only ($n = 98$)	17	81

Fisher’s exact test: $p = 0.10$, odds ratio = 1.72 (endobondic is $\sim 1.7\times$ more common among experimentally-realized compounds — the expected direction, since endobondic is meant as a kinetic-barrier signature — but not significant at this sample size).

Finer-grained (family, three-way, $n = 175$):

family	n	median Δ ICOHP/atom (eV)	% endobondic
exp_metastable	59	−1.077	27.1%
exp_stable	59	−1.589	20.3%
theo_metastable	59	−1.659	11.9%

Kruskal–Wallis across the three groups: $H = 2.81$, $p = 0.245$ — **not significant**, but the ordering is exactly the physically- motivated one: `exp_metastable` compounds (known to exist despite sitting off the convex hull — the category the endobondic/exobondic framework is specifically about) have the highest endobondic fraction; `theo_metastable` compounds (never synthesized) have the lowest.

13.3 Reading

The 7 reference reactions reproduce essentially exactly — the implementation is correct. Extending the same sign-based logic to a much larger, more heterogeneous, and generally much more comfortably stable population does not yet produce a statistically significant stable/metastable/unstable discriminator, though every trend points the right way. This is not a contradiction: the 7 reference cases were chosen specifically as borderline-stability edge cases ($\text{Pb}(\text{N}_3)_2$, $\text{S}_4\text{N}_2/\text{S}_4\text{N}_4$, ZnSn , Mn_2O_7) where a kinetic-barrier explanation is specifically needed; most of this project’s 281 compounds are not that — they are comfortably stable or comfortably not, where the thermodynamic driving force (the formation-energy correlation above) may simply dominate over the sign-based kinetic signal. **The concrete next step:** extend deliberately to more borderline-stability compounds, not just whatever the next unrelated mission happens to add to the dataset. `classify.py`’s full `ViabilityLabel` machinery (beyond the sign-only `BondingLabel` used here) needs exactly such a population — for the current dataset, decomposition into elements is strongly exothermic for nearly every compound (normal for stable inorganic chemistry), which would trivially classify almost everything `STABLE_ON_HULL` and add no information beyond what `energy_above_hull` already says directly.

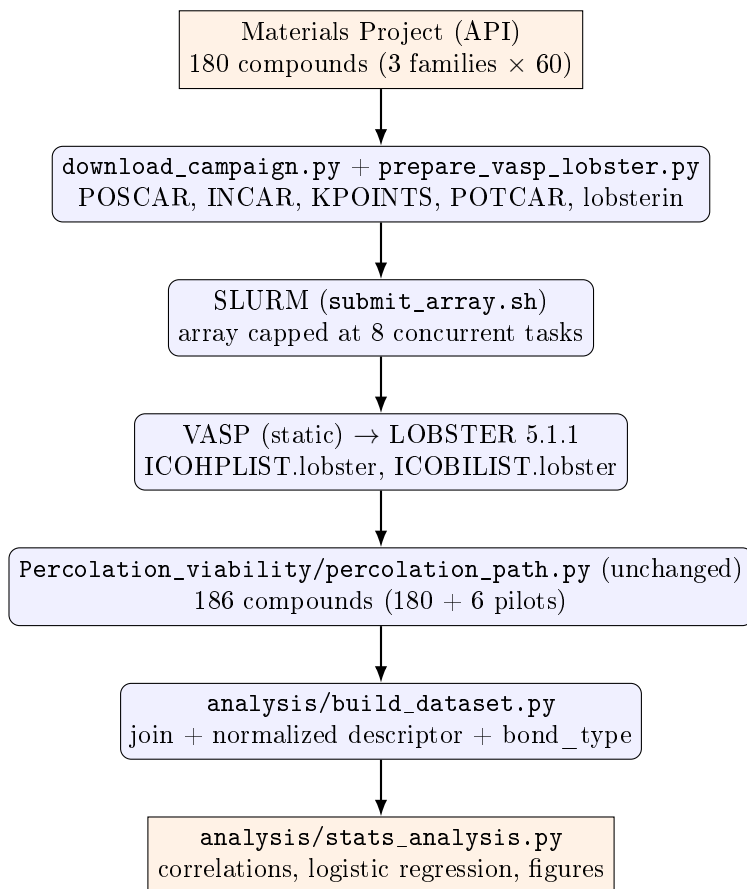
A Appendix: Percolation_viability

This appendix collects everything related to the percolation weight and its direct descendants (network dimensionality, periodic min-cut — missions #1 and #3): this was the central subject of the original campaign 2, unmodified below from its first drafting, but it is no longer this project’s organizing question (§13). The corresponding source code has been moved to the repository’s `Percolation_viability/` directory (`percolation_path.py`, `network_dimensionality.py`, `periodic_mincut.py`, and their tests `tests/test_percolation_path.py`, `tests/test_network_dimensionality.py`, `tests/test_periodic_mincut.py`).

A.1 Original objective (mission #1)

The very first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don’t already carry? Campaign 2 built a 186-compound dataset (180 new + the 6 pilot compounds) split into three families (`exp_stable`, `exp_metastable`, `theo_metastable`, §5) to settle that question.

A.2 Original workflow diagram



A.3 Campaign 2 dataset (186 compounds)

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (<code>classify()</code>)	n
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds. (Historical snapshot of campaign 2 alone; the up-to-date table across the current 349 structures is in §5.)

A.4 Results (with the percolation weight)

Group	Metric	n	ρ	p
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	0.222	0.038
ionic*	percolation weight (raw)	9	-0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

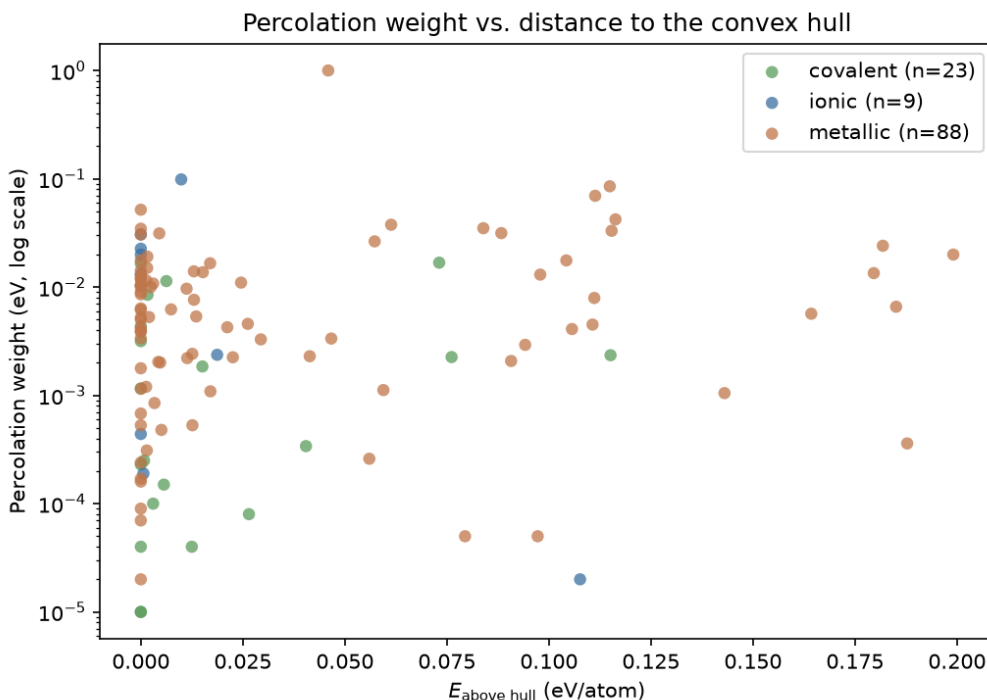
* $n < 15$: do not over-interpret.

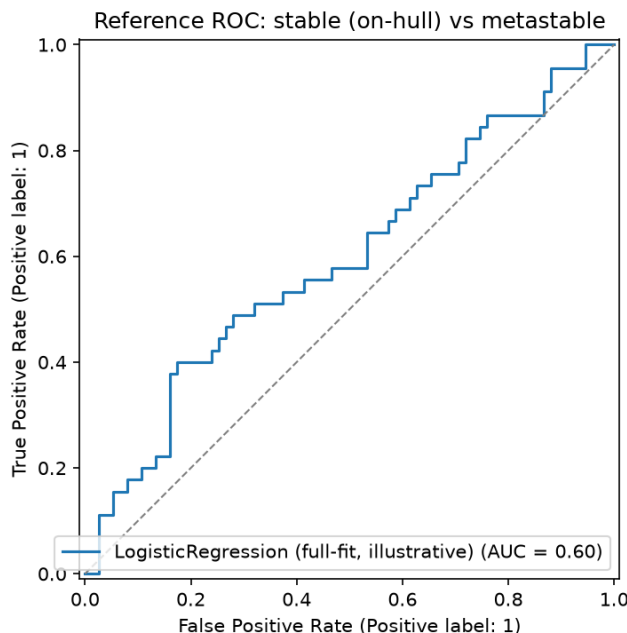
In the one subgroup where any result reaches nominal significance (metallic, $n = 88$), the normalized percolation weight ($\rho = 0.222$, $p = 0.038$) and the ICOHP sum ($\rho = 0.223$, $p = 0.037$) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset.**

Reference logistic regression. Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies ($n = 120$ after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

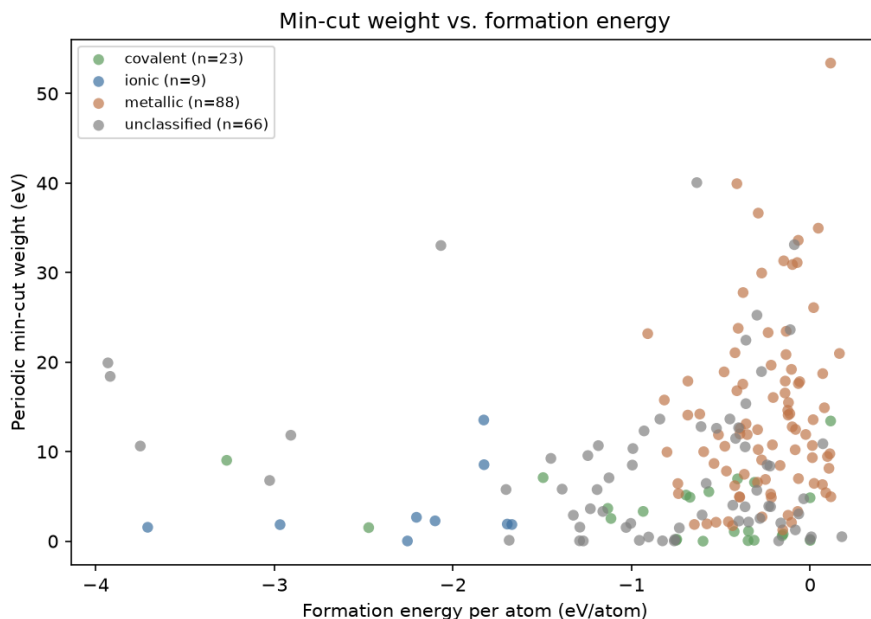
AUC ≈ 0.5 : chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.





A.5 Mission #3: network dimensionality and periodic min-cut

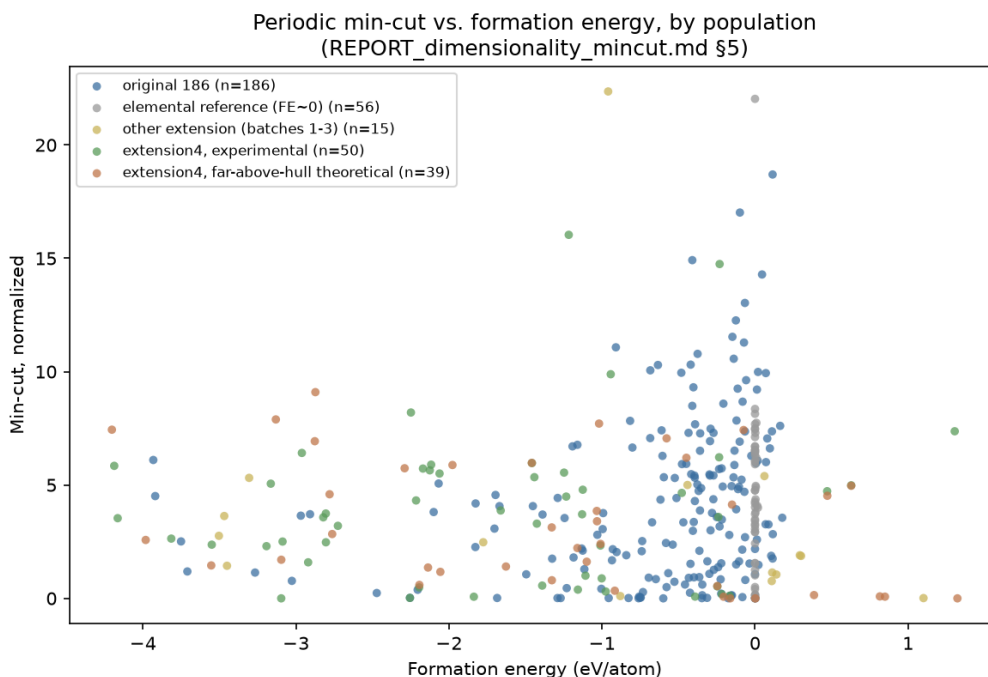
Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).



Why did the min-cut correlation collapse? Mission #3’s headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$,

essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5’s elemental references), sitting at `formation_energy_per_atom` ≈ 0 by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).
- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — a higher min-cut there is associated with a *more* negative (more stable) formation energy, the reverse of the original population’s trend.



This opposite sign is itself resolved (August-16 follow-up, four independent checks on the 39 `theo_far_hull` compounds, `analysis/investigate_theo_far_hull_signflip.py`): neither a cell-size/symmetry selection artifact (survives controlling for `n_sites+sg_number`, $\rho = -0.39$) nor a real physical effect of far-from-hull structures, but a **between-anion-chemistry confound** — F-containing systems (high min-cut, strongly negative formation energy) and azide N_3 -containing systems (near-zero min-cut, near-zero/positive formation energy) pull the pooled correlation in opposite directions. Controlling for anion identity (dummy regression, $n = 39$) or comparing the experimental/theoretical pair within the exact same chemical system ($n = 37$ pairs) both independently collapse the correlation to $\rho \approx -0.07$, not significant — the same between-group clustering failure mode this project has now hit three times (min-cut/`bond_type`, antibonding/`is_metal`, and now min-cut/`anion`).

Recommendation: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor’s) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal`.

A.6 Limits specific to the percolation weight

- **Primitive vs. conventional cell:** not reprocessed on the 186-compound dataset. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely

inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link." Tested on the 6 pilot compounds (mission #2): the choice changes the percolation weight substantially (3x-55x) but in the *opposite* direction from the hypothesis (the weight gets *smaller* with a larger cell); not extended to the full dataset based on this result.

- **Chance-level reference logistic regression** ($AUC \approx 0.5$) with percolation weight as a feature.
- **Why not SISSO**: a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space at this stage.

A.7 Conclusion of the original campaign

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. This result, together with the min-cut result (mission #3), motivated the project’s reframing around $\Delta(\text{ICOHP})$ documented in §13.

A.8 Compound List (percolation weight)

Campaign 2’s 186 compounds, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table 1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As2S3	mp-1189572	covalent	0.0000	1.00e-05
AsF3	mp-28027	covalent	0.0000	0.00e+00
BI3	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	—	0.0000	1.90e-04
BaAu2	mp-30363	metallic	0.0000	5.19e-02
BaCl2	mp-568662	ionic	0.0000	1.31e-02
BaH2	mp-23715	—	0.0000	2.40e-04
Be2Cu	mp-2031	metallic	0.0000	1.15e-03
Be2Nb3	mp-11272	metallic	0.0000	6.36e-03
Be2V	mp-11281	metallic	0.0000	6.80e-04
CaAl4	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	—	0.0000	6.00e-05
Cs2As3	mp-15557	—	0.0000	2.02e-02
FeS	mp-505531	—	0.0000	3.05e-03
Ge2Os	mp-10032	metallic	0.0000	1.21e-02
HfTc2	mp-1095669	metallic	0.0000	1.77e-02
Hg4Pt	mp-936	metallic	0.0000	3.36e-03
HgF2	mp-8177	—	0.0000	8.95e-03

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN3	mp-827	—	0.0000	2.80e-04
Li2Se	mp-2286	—	0.0000	1.16e-02
LiB	mp-1001835	—	0.0000	2.91e-03
Mg2Ni	mp-2137	metallic	0.0000	1.60e-04
MgP4	mp-384	—	0.0000	0.00e+00
MnAl12	mp-1104165	metallic	0.0000	1.42e-02
MoSe2	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	—	0.0000	0.00e+00
Na	mp-10172	—	0.0000	1.97e-03
Nb3Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa3	mp-1973	metallic	0.0000	8.58e-03
PI2	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc5Ge3	mp-17190	metallic	0.0000	5.30e-04
ScAg2	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	—	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO2	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta5Ge3	mp-17593	metallic	0.0000	3.46e-02
TaP	mp-1067587	—	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	—	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	—	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TlCl	mp-571079	—	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	—	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	—	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	—	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	—	0.0018	4.10e-04
MgH2	mp-23711	—	0.0019	1.00e-05

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	—	0.0029	7.00e-04
ClF3	mp-556767	—	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	—	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	—	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	—	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04
IF7	mp-27988	—	0.0063	0.00e+00
K	mp-10157	—	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	—	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	—	0.0118	3.50e-04
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	—	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	—	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	—	0.0269	1.93e-03
Te2Au	mp-567525	—	0.0274	1.62e-02
Be3N2	mp-6977	—	0.0315	2.00e-05
Y2O3	mp-558573	—	0.0342	5.00e-05
AgO	mp-1079720	—	0.0423	4.00e-05
FeTe2	mp-1102002	—	0.0449	5.20e-04
H2O	mp-2739191	—	0.0494	0.00e+00
CdO2	mp-2310	—	0.0562	9.00e-05
Mo3Se4	mp-21021	—	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	—	0.0800	2.06e-03
AsRh2	mp-1102364	—	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	—	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03

Theoretical, metastable ($n = 60$)

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	—	0.0033	3.59e-02
Sr3As4	mp-678007	—	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTi3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	—	0.0301	2.72e-03
Ta4C3	mp-1218000	—	0.0335	1.44e-03
Sc2O3	mp-755313	—	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTi3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	—	0.0490	5.04e-03
HCl	mp-684609	—	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	—	0.0564	8.80e-03
KSn3	mp-1185151	metallic	0.0573	2.65e-02
NbS2	mp-774873	—	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	—	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	—	0.0625	2.08e-03
ZrO2	mp-775909	—	0.0627	1.20e-04
Zn	mp-2647117	—	0.0717	1.04e-02
IN2	mp-1097011	—	0.0730	4.70e-04
CoSe2	mp-1390196	—	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	—	0.1080	4.00e-05
Na2Cl	mp-990084	—	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	—	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	—	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	—	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	—	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02

B VASP Calculation Parameters

Fixed INCAR parameters, identical across campaign 2’s 186 compounds (static, LOBSTER-compatible run) — only NBANDS (appendix D) and the k-point grid vary by compound. Same settings applied to the four **extension** batches (§12).

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (−1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMear	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (−1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen’s `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§8): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table 2: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s `MPRelaxSet` library, with one documented exception: `W` substituted for `W_pv`/`W_sv`, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)

Element	PAW-PBE potential
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)

Element	PAW-PBE potential
Zr	Zr_sv (04Jan2005)

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ -centered or Monkhorst–Pack mesh depending on cell symmetry) and NBANDS, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table 3: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154
ScAg ₂	mp-1018128	8 8 5 (G)	28

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO2	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta5Ge3	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb2	mp-11697	8 5 4 (G)	34
Te2Pd	mp-782	7 7 5 (G)	17
Te2Ru	mp-267	7 5 4 (G)	34
TeO2	mp-8377	6 5 3 (G)	48
Ti3In	mp-866046	5 5 6 (G)	72
Ti3Pt	mp-2339	5 5 5 (G)	72
TiBe12	mp-1104067	7 5 5 (G)	69
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi2	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi2	mp-11531	12 8 7 (G)	27
WO2	mp-1524394	6 5 5 (G)	68
Zr2Au	mp-2348	9 9 4 (G)	29
Zr5Pb3	mp-681992	3 3 5 (G)	154
ZrMo2	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI2	mp-567259	7 7 4 (G)	17
MgCl2	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li3Hg	mp-1646	7 7 7 (G)	24
Ta7Co6	mp-1104548	6 6 3 (G)	117
CaGa4	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH2	mp-23711	6 5 5 (G)	24
Sn7Ir5	mp-20466	4 4 4 (M)	108
NaN3	mp-1066400	7 7 7 (G)	16
ClF3	mp-556767	6 4 3 (G)	64
SiS2	mp-1095294	5 4 3 (G)	48
Sr5Sn3	mp-17720	3 3 3 (G)	104
Te2Ir	mp-1551	7 5 2 (G)	51
Li13Sn5	mp-30769	6 6 1 (G)	110
H3N	mp-643432	8 5 5 (G)	28
Cs2Se	mp-1011696	5 3 2 (G)	56
In3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb2Hg7	mp-31474	4 4 4 (G)	73
PbAu2	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO2	mp-546794	6 6 6 (M)	24
IF7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al3Os2	mp-16521	9 9 3 (G)	30

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
Nb2O5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd3	mp-7753	7 7 7 (G)	31
AsPd2	mp-1080831	8 4 4 (G)	66
GeSe2	mp-10074	5 5 5 (G)	34
Re3Ge7	mp-29141	9 6 1 (G)	180
Cu3Ge	mp-19724	6 6 5 (G)	72
Ti3Ge	mp-1189044	4 4 4 (M)	144
ZrGa3	mp-30685	7 7 5 (G)	37
SrAl2	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al12W	mp-11227	4 4 4 (M)	57
CsH	mp-632319	7 7 7 (G)	6
CaGa2	mp-992	7 7 6 (G)	23
NiHg4	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn5	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te2Au	mp-567525	7 7 5 (G)	17
Be3N2	mp-6977	10 10 3 (G)	46
Y2O3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe2	mp-1102002	4 4 4 (M)	68
H2O	mp-2739191	7 7 4 (G)	24
CdO2	mp-2310	5 5 5 (G)	68
Mo3Se4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be12Pd	mp-12666	6 6 6 (M)	69
Sr2As	mp-15697	6 6 3 (G)	28
AsRh2	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K2S	mp-559278	5 5 4 (G)	28
Ge4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li3Ag	mp-976408	7 7 7 (G)	24
Y3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr3As4	mp-678007	3 3 4 (G)	62
Ga2Se3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl2	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTi3	mp-1206636	5 5 5 (G)	32
IrCl3	mp-729507	3 3 6 (G)	84
Ta4C3	mp-1218000	6 6 6 (M)	48
Sc2O3	mp-755313	6 5 6 (G)	64
SnBr2	mp-978985	4 4 6 (M)	34
Sn7Pd5	mp-1219006	5 5 3 (G)	108
Tl4Sn	mp-1216560	5 5 5 (G)	45
LiTi3	mp-1185253	6 6 6 (M)	32

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
PtCl ₂	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K ₅ As ₄	mp-684799	4 4 3 (G)	41
KSn ₃	mp-1185151	4 4 5 (G)	64
NbS ₂	mp-774873	10 6 2 (M)	34
YAg ₃	mp-1187811	6 6 6 (G)	37
B ₃ Ir ₂	mp-1228647	9 5 4 (G)	60
TiMo ₂	mp-1216675	5 6 14 (G)	27
Cr ₂ C	mp-1226378	10 10 7 (G)	22
ZrO ₂	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN ₂	mp-1097011	5 5 5 (G)	24
CoSe ₂	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo ₃	mp-1095057	7 6 6 (G)	72
Tl ₃ Ga	mp-1187539	5 5 5 (G)	36
CaGe ₂	mp-643542	7 7 6 (G)	23
NbRh ₂	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF ₂	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na ₂ Cl	mp-990084	6 6 6 (M)	12
ZnPb ₃	mp-1187949	5 5 5 (G)	36
Sr ₃ Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al ₄ Si	mp-1228120	6 6 6 (M)	20
B ₆ Pb	mp-1096825	6 6 6 (M)	33
ScTc ₃	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd ₅	mp-1228665	6 6 6 (M)	49
Zr ₃ Be	mp-1188016	6 6 6 (M)	35
Na ₃ Cl	mp-1064484	8 8 3 (G)	16
SiAu ₃	mp-1186998	7 7 7 (G)	31
InSe ₂	mp-1232036	7 7 3 (G)	34
TiSi ₂	mp-570991	8 8 8 (M)	17
BeAu ₃	mp-983539	7 7 7 (G)	32
Zr ₃ Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg ₁₅ B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Extension Campaign: Elements and Allotropes

Every single-element structure computed across all four `extension` batches (§12; `download_extension.py` through `download_extension4.py`): the elemental decomposition reference set (`ELEMENT_REFERENCE`) plus the additional carbon allotropes (diamond, lonsdaleite, M-carbon, W-carbon) computed alongside the graphite reference specifically to test descriptors against structures whose true stable regime is elsewhere. An experimental compound’s formula is marked with a superscript star; unmarked formulas are theoretical. Auto-generated from every `mp_metadata.json` with `family=="extension"` by `report/gen_appendix_extension.py`.

Table 4: Elements and allotropes of the **extension** campaign (batches 1–4): references used to decompose multi-element compounds, plus additional carbon allotropes (TiO₂’s polymorphs are a binary compound, see the compounds table instead). * = experimental reference (unmarked = theoretical). No Δ ICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6 ₃ /mmc	0.0000
Br*	mp-23154	Cmce	0.0000
C	mp-1080826	C2/m	0.3009
C	mp-1190171	Pnma	0.2927
C*	mp-66	Fd-3m	0.1123
C*	mp-48	P6 ₃ /mmc	0.0031
C*	mp-47	P6 ₃ /mmc	0.1395
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6 ₃ /mmc	0.0000
Cl2*	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F2*	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H2*	mp-730101	P2 _{12_12_1}	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.0000
Hg*	mp-10861	P6 ₃ /mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4 ₁ /mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N2	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O2*	mp-1524462	C2/m	0.0000
Os*	mp-49	P6 ₃ /mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6 ₃ /mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6 ₃ /mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Sc*	mp-67	P6_3/mmc	0.0000
Se*	mp-14	P3_121	0.0011
Sn*	mp-623511	I4/mmm	0.0614
Sr*	mp-139	P6_3/mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6_3/mmc	0.0000
Te*	mp-19	P3_121	0.0000
Ti*	mp-46	P6_3/mmc	0.0152
Tl*	mp-82	P6_3/mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6_3/mmc	0.0000
Zn*	mp-79	P6_3/mmc	0.0000
Zr*	mp-131	P6_3/mmc	0.0000

F Extension Campaign: Compounds and Polymorphs

Every multi-element target compound across all four `extension` batches (100 total, 89 from `extension4`) — including the TiO2 high-pressure polymorphs (`extension2`) alongside every `extension4` alkali/alkaline-earth binary — with Δ ICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icoh` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension.py`.

Table 5: Target compounds and polymorphs of the `extension` campaign (batches 1–4, 163 compounds, 89 from `extension4`) – includes TiO2’s high-pressure polymorphs and every multi-element compound from earlier batches. * = experimental compound (unmarked = theoretical). Δ ICOHP per atom, products – reactants convention (`reaction_analysis`); “–” where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	Δ ICOHP/at. (eV)	Label
Ba5P4*	mp-17566	Pnma	0.0000	-0.9590	exobondic
BaCl2*	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaF3*	mp-1080821	P2_1/m	0.0000	-0.8680	exobondic
BaF3	mp-1183303	I4/mmm	0.0612	-0.2493	exobondic
BaN2*	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaO*	mp-1008500	P6_3/mmc	0.0189	-0.3221	exobondic
BaO	mp-1950989	P6_3/mmc	0.0434	-1.6129	exobondic
BaS3*	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
Be3N2*	mp-18337	Ia-3	0.0000	-0.0746	exobondic
Be3N2	mp-1017550	P-3m1	0.2081	0.3960	endobondic
Be3P2*	mp-567841	Ia-3	0.0000	-0.2111	exobondic
Be3P2	mp-1084771	Pn-3m	0.1567	-0.1795	exobondic
BeCl2*	mp-23267	Ibam	0.0087	1.8204	endobondic
BeCl2	mp-1172909	I-43m	3.1688	-5.7807	exobondic
BeF2*	mp-15951	P3_121	0.0004	3.1514	endobondic
BeF2	mp-975644	P6/mmm	2.5181	-0.6816	exobondic
BeO*	mp-7599	P4_2/mnm	0.0153	2.7864	endobondic
BeO	mp-1778	F-43m	0.0070	4.4391	endobondic
Ca3N2*	mp-1047	R-3c	0.0167	-4.4881	exobondic
Ca3N2*	mp-844	Ia-3	0.0000	-4.4374	exobondic
Ca3N2	mp-1013524	Pm-3m	0.5112	-4.4898	exobondic
CaCl2*	mp-23214	Pnnm	0.0011	-0.3199	exobondic

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	Δ ICOHP/at. (eV)	Label
CaCl ₂	mp-1120739	Fm-3m	0.0422	-0.2208	exobondic
CaF ₂ *	mp-10464	Pnma	0.0346	-0.2227	exobondic
CaF ₂	mp-554355	P4/mmm	0.2403	-0.1814	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
CaO*	mp-1079707	Cmce	0.2098	-2.2127	exobondic
CaO*	mp-2605	Fm-3m	0.0000	-0.9525	exobondic
CaO	mp-1181903	Fd-3m	2.7281	-2.9971	exobondic
Cs ₂ S*	mp-1077814	P6 ₃ /mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
CsN ₃ *	mp-510557	I4/mcm	0.0000	-2.2903	exobondic
CsN ₃	mp-1225892	P4/mmm	0.0113	-1.6664	exobondic
CsO ₂ *	mp-1441	I4/mmm	0.0349	0.0666	endobondic
CsO ₂	mp-1096936	Cmmm	0.0940	0.7743	endobondic
CsP ₇ *	mp-1201790	Pbcm	0.0000	-0.2978	exobondic
CsP ₇	mp-1213206	Pca2 ₁	0.0001	-0.2667	exobondic
K ₂ O*	mp-971	Fm-3m	0.0000	-0.4357	exobondic
K ₂ O	mp-1223784	P4/mmm	0.2000	-1.5638	exobondic
K ₂ S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KF ₃ *	mp-1544557	P2 ₁₂₁₂₁	0.0125	-1.1511	exobondic
KF ₃	mp-2749823	P2 ₁	0.0119	-1.1505	exobondic
KN ₃	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
KP*	mp-1058315	R-3m	1.0080	-2.1600	exobondic
KP	mp-1057787	Immm	1.0092	-2.1366	exobondic
Li ₂ O*	mp-1960	Fm-3m	0.0000	2.3366	endobondic
Li ₂ O	mp-755894	Pnma	0.0844	1.9481	endobondic
Li ₂ S*	mp-1125	Pnma	0.0615	1.8505	endobondic
Li ₂ S	mp-557142	P6 ₃ /mmc	0.1750	1.5052	endobondic
Li ₃ N*	mp-2341	P6 ₃ /mmc	0.0039	0.6165	endobondic
LiCl*	mp-22905	Fm-3m	0.0243	2.8030	endobondic
LiCl	mp-1185319	P6 ₃ mc	0.0039	2.5005	endobondic
LiF*	mp-1138	Fm-3m	0.0000	4.4673	endobondic
LiF	mp-1009009	Pm-3m	0.2875	4.5131	endobondic
LiP ₅ *	mp-2412	Pna2 ₁	0.0077	0.9043	endobondic
LiP ₅	mp-32760	Pna2 ₁	0.0963	0.8692	endobondic
MgCl ₂ *	mp-23210	R-3m	0.0000	0.1568	endobondic
MgCl ₂	mp-569626	Ama2	0.1991	-0.0862	exobondic
MgF ₂ *	mp-1072956	Pnnm	0.0025	0.0015	endobondic
MgF ₂	mp-1062842	Amm2	0.2636	0.0229	endobondic
MgO*	mp-549706	P6 ₃ mc	0.1180	-1.4179	exobondic
MgO	mp-1009127	Pm-3m	0.7484	-1.0763	exobondic
MgS*	mp-1315	Fm-3m	0.0000	-0.3401	exobondic
MgS	mp-13032	F-43m	0.0343	-0.4163	exobondic
Mn ₂ O ₇ *	mp-28338	P2 ₁ /c	0.3346	-1.4687	exobondic
MnO ₂ *	mp-510408	P4 ₂ /mnm	0.0464	-0.5274	exobondic
Na ₂ Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
Na ₂ O*	mp-2352	Fm-3m	0.0000	-1.3057	exobondic
Na ₂ O	mp-1975297	C2/m	0.0981	-1.5619	exobondic
NaN ₃ *	mp-22003	R-3m	0.0017	-1.5201	exobondic
NaN ₃	mp-1065265	C2/m	1.0162	-5.4248	exobondic
NaS*	mp-409	P-62m	0.0023	-0.4911	exobondic
NaS	mp-1180106	C2/m	1.0061	-2.3999	exobondic
PbN ₆ *	mp-667338	Pnma	0.1093	-2.5257	exobondic
Rb ₂ O*	mp-1394	Fm-3m	0.0032	0.7235	endobondic
Rb ₂ O	mp-2023482	P-3m1	0.0479	-0.7592	exobondic

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)	Label
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
RbF*	mp-2064	Pm-3m	0.0516	3.9244	endobondic
RbF	mp-11718	Fm-3m	0.0000	1.9497	endobondic
RbN3*	mp-581833	P4/mmm	0.0193	-1.3478	exobondic
RbN3	mp-1219567	Amm2	0.7966	-3.5636	exobondic
RbP*	mp-1179688	C2/m	0.9879	-1.9315	exobondic
RbP	mp-1058499	Immm	0.9882	-1.9104	exobondic
RbS*	mp-558071	Immm	0.0019	-0.3217	exobondic
RbS	mp-1179684	C2/m	0.9975	-2.7347	exobondic
S2N*	COD:4031496	P4_2nm	—	-0.9767	exobondic
SN*	COD:7017102	P2_1/c	—	-1.3963	exobondic
Sr3P4*	mp-14288	Fdd2	0.0000	-1.0472	exobondic
Sr3P4	mp-682953	P1	0.0081	-1.0793	exobondic
SrF2*	mp-1102240	Pnma	0.0646	-0.5421	exobondic
SrF2	mp-1102911	Pnma	0.0267	-0.1481	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrO2*	mp-726705	Pnma	0.0096	-1.9205	exobondic
SrO2	mp-2763179	Pnma	0.0110	-1.7522	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic
TiO2*	mp-1439	Pbcn	0.0045	-0.6461	exobondic
TiO2*	mp-9173	Pnma	0.0575	-0.6509	exobondic
TiO2*	mp-430	P2_1/c	0.0395	-0.6267	exobondic

G Extension Campaign: Endobondic Reactions

The subset of case-1 decomposition reactions ($A_a B_b \rightarrow a A + b B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products — reactants convention — note this is the sign *opposite* `reaction_icohp.py`’s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`’s docstring).

Table 6: Decomposition-into-elements (case 1) reactions labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products — reactants) among the 163 compounds of the **extension** campaign. * = experimental starting compound.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
0.3333 Be3N2 $\rightarrow$ Be + 0.3333 N2	mp-1017550	0.2081	0.3960
BeCl2* $\rightarrow$ Be + Cl2	mp-23267	0.0087	1.8204
BeF2* $\rightarrow$ Be + F2	mp-15951	0.0004	3.1514
BeO* $\rightarrow$ Be + 0.5 O2	mp-7599	0.0153	2.7864
BeO $\rightarrow$ Be + 0.5 O2	mp-1778	0.0070	4.4391
0.5 Cs2S* $\rightarrow$ Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl* $\rightarrow$ Cs + 0.5 Cl2	mp-573697	0.0071	0.8830
CsF* $\rightarrow$ Cs + 0.5 F2	mp-8455	0.1088	3.8464
CsO2* $\rightarrow$ Cs + O2	mp-1441	0.0349	0.0666
CsO2 $\rightarrow$ Cs + O2	mp-1096936	0.0940	0.7743
KCl* $\rightarrow$ K + 0.5 Cl2	mp-23289	0.0776	0.7616
0.5 Li2O* $\rightarrow$ Li + 0.25 O2	mp-1960	0.0000	2.3366
0.5 Li2O $\rightarrow$ Li + 0.25 O2	mp-755894	0.0844	1.9481
0.5 Li2S* $\rightarrow$ Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li2S $\rightarrow$ Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li3N* $\rightarrow$ Li + 0.1667 N2	mp-2341	0.0039	0.6165
LiCl* $\rightarrow$ Li + 0.5 Cl2	mp-22905	0.0243	2.8030
LiCl $\rightarrow$ Li + 0.5 Cl2	mp-1185319	0.0039	2.5005

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{LiF}^* \rightarrow \text{Li} + 0.5 \text{ F}_2$	mp-1138	0.0000	4.4673
$\text{LiF} \rightarrow \text{Li} + 0.5 \text{ F}_2$	mp-1009009	0.2875	4.5131
$\text{LiP}_5^* \rightarrow \text{Li} + 5 \text{ P}$	mp-2412	0.0077	0.9043
$\text{LiP}_5 \rightarrow \text{Li} + 5 \text{ P}$	mp-32760	0.0963	0.8692
$\text{MgCl}_2^* \rightarrow \text{Mg} + \text{Cl}_2$	mp-23210	0.0000	0.1568
$\text{MgF}_2^* \rightarrow \text{Mg} + \text{F}_2$	mp-1072956	0.0025	0.0015
$\text{MgF}_2 \rightarrow \text{Mg} + \text{F}_2$	mp-1062842	0.2636	0.0229
$0.5 \text{ Rb}_2\text{O}^* \rightarrow \text{Rb} + 0.25 \text{ O}_2$	mp-1394	0.0032	0.7235
$\text{RbCl}^* \rightarrow \text{Rb} + 0.5 \text{ Cl}_2$	mp-23299	0.0298	1.7924
$\text{RbF}^* \rightarrow \text{Rb} + 0.5 \text{ F}_2$	mp-2064	0.0516	3.9244
$\text{RbF} \rightarrow \text{Rb} + 0.5 \text{ F}_2$	mp-11718	0.0000	1.9497

H Extension Campaign: Exobondic Reactions

The complementary **exobondic** subset ($\Delta\text{ICOHP} < 0$), same convention and provenance as Appendix G.

Table 7: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta\text{ICOHP} < 0$, products – reactants) among the 163 compounds of the **extension** campaign. * = experimental starting compound.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$0.2 \text{ Ba}_5\text{P}_4^* \rightarrow \text{Ba} + 0.8 \text{ P}$	mp-17566	0.0000	-0.9590
$\text{BaCl}_2^* \rightarrow \text{Ba} + \text{Cl}_2$	mp-23199	0.0068	-0.2244
$\text{BaF}_3^* \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1080821	0.0000	-0.8680
$\text{BaF}_3 \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1183303	0.0612	-0.2493
$\text{BaN}_2^* \rightarrow \text{Ba} + \text{N}_2$	mp-1102371	2.0568	-8.6248
$\text{BaO}^* \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1008500	0.0189	-0.3221
$\text{BaO} \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1950989	0.0434	-1.6129
$\text{BaS}_3^* \rightarrow \text{Ba} + 3 \text{ S}$	mp-556296	0.0436	-0.6233
$0.3333 \text{ Be}_3\text{N}_2^* \rightarrow \text{Be} + 0.3333 \text{ N}_2$	mp-18337	0.0000	-0.0746
$0.3333 \text{ Be}_3\text{P}_2^* \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-567841	0.0000	-0.2111
$0.3333 \text{ Be}_3\text{P}_2 \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-1084771	0.1567	-0.1795
$\text{BeCl}_2 \rightarrow \text{Be} + \text{Cl}_2$	mp-1172909	3.1688	-5.7807
$\text{BeF}_2 \rightarrow \text{Be} + \text{F}_2$	mp-975644	2.5181	-0.6816
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1047	0.0167	-4.4881
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-844	0.0000	-4.4374
$0.3333 \text{ Ca}_3\text{N}_2 \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1013524	0.5112	-4.4898
$\text{CaCl}_2^* \rightarrow \text{Ca} + \text{Cl}_2$	mp-23214	0.0011	-0.3199
$\text{CaCl}_2 \rightarrow \text{Ca} + \text{Cl}_2$	mp-1120739	0.0422	-0.2208
$\text{CaF}_2^* \rightarrow \text{Ca} + \text{F}_2$	mp-10464	0.0346	-0.2227
$\text{CaF}_2 \rightarrow \text{Ca} + \text{F}_2$	mp-554355	0.2403	-0.1814
$\text{CaN}^* \rightarrow \text{Ca} + 0.5 \text{ N}_2$	mp-1058549	0.3982	-4.9183
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1079707	0.2098	-2.2127
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-2605	0.0000	-0.9525
$\text{CaO} \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1181903	2.7281	-2.9971
$\text{CsN}_3^* \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-510557	0.0000	-2.2903
$\text{CsN}_3 \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-1225892	0.0113	-1.6664
$\text{CsP}_7^* \rightarrow \text{Cs} + 7 \text{ P}$	mp-1201790	0.0000	-0.2978
$\text{CsP}_7 \rightarrow \text{Cs} + 7 \text{ P}$	mp-1213206	0.0001	-0.2667
$0.5 \text{ K}_2\text{O}^* \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-971	0.0000	-0.4357
$0.5 \text{ K}_2\text{O} \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-1223784	0.2000	-1.5638
$0.5 \text{ K}_2\text{S}^* \rightarrow \text{K} + 0.5 \text{ S}$	mp-1022	0.0000	-0.3359

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	Δ ICOHP/at. (eV)
KF3* $\rightarrow$ K + 1.5 F2	mp-1544557	0.0125	-1.1511
KF3 $\rightarrow$ K + 1.5 F2	mp-2749823	0.0119	-1.1505
KN3 $\rightarrow$ K + 1.5 N2	mp-636056	1.0370	-5.9991
KP* $\rightarrow$ K + P	mp-1058315	1.0080	-2.1600
KP $\rightarrow$ K + P	mp-1057787	1.0092	-2.1366
MgCl2 $\rightarrow$ Mg + Cl2	mp-569626	0.1991	-0.0862
MgO* $\rightarrow$ Mg + 0.5 O2	mp-549706	0.1180	-1.4179
MgO $\rightarrow$ Mg + 0.5 O2	mp-1009127	0.7484	-1.0763
MgS* $\rightarrow$ Mg + S	mp-1315	0.0000	-0.3401
MgS $\rightarrow$ Mg + S	mp-13032	0.0343	-0.4163
0.5 Mn2O7* $\rightarrow$ Mn + 1.75 O2	mp-28338	0.3346	-1.4687
MnO2* $\rightarrow$ Mn + O2	mp-510408	0.0464	-0.5274
0.5 Na2Cl $\rightarrow$ Na + 0.25 Cl2	mp-1077015	0.2391	-0.3965
0.5 Na2O* $\rightarrow$ Na + 0.25 O2	mp-2352	0.0000	-1.3057
0.5 Na2O $\rightarrow$ Na + 0.25 O2	mp-1975297	0.0981	-1.5619
NaN3* $\rightarrow$ Na + 1.5 N2	mp-22003	0.0017	-1.5201
NaN3 $\rightarrow$ Na + 1.5 N2	mp-1065265	1.0162	-5.4248
NaS* $\rightarrow$ Na + S	mp-409	0.0023	-0.4911
NaS $\rightarrow$ Na + S	mp-1180106	1.0061	-2.3999
PbN6* $\rightarrow$ Pb + 3 N2	mp-667338	0.1093	-2.5257
0.5 Rb2O $\rightarrow$ Rb + 0.25 O2	mp-2023482	0.0479	-0.7592
RbN3* $\rightarrow$ Rb + 1.5 N2	mp-581833	0.0193	-1.3478
RbN3 $\rightarrow$ Rb + 1.5 N2	mp-1219567	0.7966	-3.5636
RbP* $\rightarrow$ Rb + P	mp-1179688	0.9879	-1.9315
RbP $\rightarrow$ Rb + P	mp-1058499	0.9882	-1.9104
RbS* $\rightarrow$ Rb + S	mp-558071	0.0019	-0.3217
RbS $\rightarrow$ Rb + S	mp-1179684	0.9975	-2.7347
0.5 S2N* $\rightarrow$ S + 0.25 N2	COD:4031496	—	-0.9767
SN* $\rightarrow$ S + 0.5 N2	COD:7017102	—	-1.3963
0.3333 Sr3P4* $\rightarrow$ Sr + 1.333 P	mp-14288	0.0000	-1.0472
0.3333 Sr3P4 $\rightarrow$ Sr + 1.333 P	mp-682953	0.0081	-1.0793
SrF2* $\rightarrow$ Sr + F2	mp-1102240	0.0646	-0.5421
SrF2 $\rightarrow$ Sr + F2	mp-1102911	0.0267	-0.1481
SrN* $\rightarrow$ Sr + 0.5 N2	mp-1058586	0.4343	-4.5395
SrO2* $\rightarrow$ Sr + O2	mp-726705	0.0096	-1.9205
SrO2 $\rightarrow$ Sr + O2	mp-2763179	0.0110	-1.7522
SrS* $\rightarrow$ Sr + S	mp-10627	0.2996	-0.1088
TiO2* $\rightarrow$ Ti + O2	mp-1439	0.0045	-0.6461
TiO2* $\rightarrow$ Ti + O2	mp-9173	0.0575	-0.6509
TiO2* $\rightarrow$ Ti + O2	mp-430	0.0395	-0.6267